Over the past five years of my PhD, I have focused on integrating data science solutions into clinical systems through collaborative research with key domain stakeholders, such as consumers, genetic counselors, and physicians. By prioritizing stakeholder perspectives, I have developed more usable and effective solutions to address healthcare delivery challenges. In the process of this research, I have grown increasingly interested in how artificial intelligence (AI) tools have influenced the lives of both patients and providers, shaping the ways patients receive care and how providers practice. As generative AI is further embedded into our collective consciousness (and in many ways, becomes a new collaborator in our lives), it is imperative that we thoughtfully assess tools to mitigate potential harm and maximize public benefit.

My previous research has enabled me to become deeply acquainted with the engineering challenges of implementing natural language processing tools within secure healthcare systems and granted me insights into factors that influence clinicians' and consumers' acceptance of generative AI agents. However, this work does not fully uncover how humans interact with the AI tools and the underlying reasoning guiding their interactions. As AI systems are increasingly leveraged as reasoning partners and collaborators, I believe that I must also gain a deeper understanding of human reasoning as compared to the reasoning paradigms we have built into generative AI models. This knowledge is critical for investigating human-AI collaboration and will lead to the design of more human-aligned AI systems. Therefore, I am interested in pursuing a distinct pivot toward computational cognitive science, studying models of human reasoning and building a stronger theoretical perspective on AI-human interaction more broadly. The Schmidt Sciences Fellowship would grant me the opportunity to pursue this research and build towards my long-term goals.

If I were granted the fellowship, I would explore how to use Bayesian models to represent knowledge-dependent human reasoning and develop human-AI collaboration frameworks. Currently, such frameworks often use Bayesian probabilistic models as a foundation for representing collaborative thought.[1] Advances in these systems have achieved state-of-the-art performance by treating users as cooperative planners and scoring ambiguous instructions via large-language-model likelihoods.[2] However, these systems assume that all users have complete and identical domain knowledge. In real life scenarios, we know that information literacy can influence decision-making, and in domains such as healthcare, this could translate into disparate outcomes for groups with groups lower literacy.[3] Additionally, when collaborating with an AI system, misinterpretations may arise when non-experts provide suboptimal, uninformed commands or experts use overly context-rich language that the assistant cannot parse correctly. Therefore, it is important to understand how levels of user knowledge shape their interactions with AI assistants and decisions towards a goal.

To address this challenge, I propose extending Bayesian inverse planning with the inclusion of a novel latent *knowledge level* variable K that conditions both the user's goal (G) prior $P(G \mid K)$ and their action-language instruction likelihoods by completing the following specific aims:

In Aim 1, I propose developing and implementing a joint goal–knowledge Bayesian inverse planning framework. By jointly inferring a user's goal (*G)* and their knowledge (*K)*, this framework aims to be knowledge level adaptive, providing extra guidance when *K* is low and deferring when *K* is high. This builds on Bayesian Theory-of-Mind approaches that infer agents' beliefs and goals under uncertainty.[4] In Aim 2, I will conduct user studies to test the model's fit and its effect on collaboration metrics. I will construct a validation study with users and experimentally varying participants' background information (high vs. low levels of information) for a certain task. We will fit the joint *(G, K)* inference model to their instructions to an AI assistant and actions. The hypothesis is that this joint inference model will improve upon simpler systems only considering *G*, assisting participants based on the inferred knowledge level. In Aim 3, I will build and evaluate a real-time knowledge-adaptive assistant across multiple tasks. Building on these insights from Aims 1 and 2, I will implement a knowledge-adaptive Assistant that tailors clarifications and guidance dynamically based on the inferred *K*. Inspired by educational AI, where modeling user misunderstandings via Bayesian inverse planning enabled personalized feedback and improved learning outcomes, I anticipate similar benefits in collaborative settings, with low knowledge users receiving additional support and providing expert users with ease of interaction.[5]

To effectively execute this proposal, I aim to join a computational cognitive science host lab with a research environment valuing curiosity and openness, as I would likely be bringing a different domain perspective than some of their lab members. In my experience, I have benefited greatly from cross-disciplinary collaborations and would likewise seek out a group that encourages such collaborations within their lab, across academic institutions, and perhaps beyond. This would be critical in achieving my goals to expand my research network and learn from diverse points of view. I would be interested in joining the lab of Dr. Joshua B. Tenenbaum (MIT Computational Cognitive Science group), whose lab has pioneered Bayesian Theory of Mind framework and the Cooperative Language-Guided Inverse Planning design that this proposal builds upon, or Dr. Tom Griffiths (Princeton Computational Cognitive Science Lab) whose work on nonparametric Bayesian modeling may expand the current limited parametric representation of this proposal. Likewise, Dr. Chris Lucas's expertise (Lucas Lab at University of Edinburgh) in inductive reasoning would provide great help in the incorporation of knowledge as an inductive prior in the proposed model.

The proposed work would advance cognitive science by formalizing how knowledge uncertainty influences human planning and communication, improve AI assistance by enabling transparent, personalized support that adapts to user expertise, and lay the groundwork for effective human-AI systems with applications to many societal domains. Completing this proposed Schmidt Sciences Fellowship research would support my long-term career aspirations to contribute to AI alignment and human-AI interaction science.

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